

Parallel Consensus Routing: Training-Free Quality Control, Coordinated Hallucination, and Model-Pair Selection in Cost-Efficient LLM Inference

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Abstract—Cost-efficient large language model (LLM) inference is dominated by two paradigms: *pre-query routing*, which commits to a model before any output exists, and *sequential cascading*, which escalates cheap-to-expensive in series and accrues additive latency. We study Parallel Consensus Routing (PCR), a third paradigm that runs two inexpensive models simultaneously and uses their binary output agreement as a training-free quality oracle: agreed queries are resolved immediately, disagreements trigger exactly one frontier escalation. Extending an initial single-pair study (MMLU: 92.7% accuracy at 27.5% escalation with domain-calibrated thresholds), we run a controlled matrix of eleven cheap-model pairs across seven benchmarks (MMLU, GSM8K, ARC-Challenge, OpenBookQA, CommonsenseQA, TruthfulQA, MMLU-Pro) and score eight routing policies on identical stored data. Five results emerge. (1) Coordinated hallucination (C-Hall)—two cheap models agreeing on a wrong answer—scales monotonically with question difficulty, not task type: from 2% on OpenBookQA to $\approx 30\%$ on MMLU-Pro, reaching mathematical-reasoning levels on a plain multiple-choice task. (2) A capability-matched cheap pair with PCR matches always-frontier accuracy on easy/medium questions at 54–69% lower cost, but on hard questions its accuracy is safe while its C-Hall (20–27%) is not. (3) Model-family relatedness raises agreement rate substantially (5/6 benchmarks, bootstrap CI excludes 0) but its effect on C-Hall is at most marginal—every knowledge-benchmark CI spans zero; only GSM8K shows a within-family gap (+6.8 pp), and that is a single pair. The agreement effect is robust; a coordinated-hallucination effect is not established. The binding methodological constraint is capability matching, not family. (4) A third cheap model (PCR-3, majority vote) reliably breaks ties but yields no accuracy or safety gain. (5) On free-tier APIs, two of {gpt-oss-20B, Qwen3.6-27B, Qwen3.8-27B} on a low-latency provider is the recommended default: solo accuracies within ~ 1 pp, ≈ 0.5 s latency, ≈ 5 –12% escalation. All experiments used free-tier inference and required \$0 marginal API spend.

Index Terms—LLM routing, cost-efficient inference, parallel consensus, coordinated hallucination, model-pair selection, capability matching, black-box API inference

I. INTRODUCTION

The LLM API marketplace is sharply tiered: frontier models command costs one to two orders of magnitude above small open models [1]. For throughput workloads, routing *all* queries to a frontier is prohibitively expensive and routing all to a cheap model forfeits quality on the hard tail. The engineering problem is: *given a query, which model or combination minimizes cost subject to a quality constraint?*

Three paradigms exist. **Pre-query routing** [2], [3] predicts the right tier from input features before any output is generated. **Sequential cascading** [4] queries cheap first and applies a learned scorer to decide escalation, adding one model’s latency per hop. **Self-verification routing** [7] lets a small model grade its own answer, but self-verification is noisy on reasoning [13].

We study a structurally different idea: **use two cheap models as quality oracles for each other**. Two independently deployed cheap models queried in parallel that converge on the same answer provide agreement evidence that is stronger than either model’s self-assessment. On disagreement, exactly one frontier call is made. The system is training-free, black-box compatible, and faster than a serial cascade on every agreed query. The analogy to speculative decoding [5], [6] is exact—draft-then-verify at the response level rather than the token level—but without the white-box probability access speculative decoding requires.

The failure mode that makes this non-trivial is **coordinated hallucination (C-Hall)**: two cheap models agreeing on the *same wrong* answer. C-Hall is invisible to the router—it is indistinguishable from correct consensus—so understanding when it occurs is a safety prerequisite, not an optimization.

a) Contributions: This paper (i) reproduces and extends an initial single-pair PCR study with a **controlled 11-pair, 7-benchmark matrix**; (ii) shows C-Hall **scales with difficulty, not task type**, generalizing the earlier “knowledge vs. math” dichotomy to a difficulty-tier one; (iii) runs a **full 8-policy ablation** (Always-Cheap, Best-Cheap, Consensus, PCR-2, Oracle-2-cheap, Random, Always-Frontier, PCR-3) on identical data; (iv) resolves the **model-family question**—agreement effect real, C-Hall effect absent—and identifies **capability matching** as the binding methodological constraint; (v) provides measured cost/latency and a concrete **practitioner pair-selection guide**.

II. RELATED WORK

A. Sequential Cascade Routing

FrugalGPT [4] defines an LLM cascade with a learned (DistilBERT) accept/escalate scorer, matching GPT-4 at up to 98% cost reduction on classification, but requiring task-specific supervised training and incurring latency proportional to cascade depth. Hybrid LLM [3] predicts the small-vs-large

quality gap pre-query (up to 40% fewer large calls, <1% quality drop) but cannot use observed output. AutoMix [7] is architecturally closest to PCR: a single model self-verifies before escalation, with a POMDP meta-verifier absorbing self-verification noise. Our cross-model agreement is stronger than self-verification in principle—independent models cannot share a self-delusion—though, as our matrix shows, shared training corpora still induce correlated errors.

B. Pre-Query Routing

RouteLLM [2] trains four router variants on Chatbot Arena preference data (>85% cost reduction on MT-Bench, 45% on MMLU). GraphRouter [25] models query-LLM relationships with GNNs; FORC [24] learns per-LLM score prediction via contrastive learning; RouterBench [21] standardizes evaluation. All require trained components; PCR requires none.

C. Multi-Model Consensus and Ensembles

Multi-agent debate [8] improves reasoning through iterative text exchange at $O(\text{rounds} \times \text{models})$ cost. Self-Consistency [10] majority-votes multiple samples from one model; LLM-Blender [9] fuses open-model outputs with a trained ranker; ChatEval [26] debates for evaluation. None exploit cost differentials between tiers, and none use a single-round binary agreement signal.

D. Speculative Decoding

Leviathan et al. [5] and Chen et al. [6] accelerate decoding with a draft model whose tokens the target verifies by rejection sampling ($2\text{--}3\times$ speedup, provable equivalence). This needs the target’s per-token distribution—unavailable for proprietary APIs. PCR realizes the same draft-then-verify separation at the *response* level on final text only.

E. Semantic Caching, Calibration, and Hallucination

vCache [11] shows global similarity thresholds carry no correctness guarantee and proposes per-embedding online learning; our per-domain/per-difficulty threshold table is the routing analogue. SelfCheckGPT [12] detects hallucination via multi-sample consistency; Kadavath et al. [13] show LLMs are poorly calibrated on multi-step reasoning. Coordinated hallucination—agreement *negatively* correlated with correctness on hard items—has not been characterized in the routing literature.

F. Benchmarks

MMLU [14], GSM8K [15], and ARC-Challenge [16] are standard. We add TruthfulQA [17] (adversarial misconceptions), MMLU-Pro [18] (10-option, reasoning-heavy), OpenBookQA [19], and CommonsenseQA [20]. Chain-of-thought [23] is suppressed to isolate the agreement signal.

G. Research Gap

No prior work evaluates a *parallel binary-consensus* quality oracle for cost reduction, characterizes coordinated hallucination and its difficulty-dependence, controls for model-family relatedness while holding capability constant, or provides an empirical pair-selection guide for practitioners. This paper does all four.

III. METHODOLOGY

A. Problem Formulation

Let M_A, M_B be cheap models and M_F a frontier model, with $\text{cost}(M_F) \gg \text{cost}(M_A) \approx \text{cost}(M_B)$. For a multiple-choice query q each model emits a letter $r \in \{A, \dots\}$. The **PCR-2** decision rule is

$$\text{output}(q) = \begin{cases} r_A(q) & r_A(q) = r_B(q) \quad [\text{agreement}] \\ M_F(q) & r_A(q) \neq r_B(q) \quad [\text{disagreement}] \end{cases} \quad (1)$$

with expected per-query cost $\mathbb{E}[\text{cost}] = \text{cost}(M_A) + \text{cost}(M_B) + \text{cost}(M_F) \cdot P(\text{disagree})$.

B. PCR-3: Majority Vote

PCR-3 runs three cheap models M_A, M_B, M_C ; if any two agree, their answer is taken; only if all three differ is M_F called. This trades one extra cheap call for a lower escalation rate and a majority-vote margin.

C. Policy Space

Every policy below is scored on the *same* stored per-question data (cheap answers, agreement, frontier answer on disagreements, per-call token and latency instrumentation), so no policy requires additional API calls except Always-Frontier (see §IV): *Always-Cheap-A/B* (one small model); *Best-Cheap* (the stronger of the two); *Consensus, no escalation* (trust agreement, fall back to M_A on disagreement—provably equal to Always-Cheap-A); *PCR-2*; *Oracle-2-cheap* (correct whenever *either* cheap model is correct—the ceiling for any two-cheap method); *Random* (coin flip M_A vs. M_F); *Always-Frontier*; *PCR-3*.

D. Agreement Signal and Outcome Categories

We optionally compute embedding cosine similarity $s \in [-1, 1]$ between raw responses (all-MiniLM-L6-v2 [22]). Three outcome categories: **correct consensus** ($r_A = r_B = \text{gold}$); **coordinated hallucination** ($r_A = r_B \neq \text{gold}$); **disagreement** ($r_A \neq r_B$).

E. Coordinated Hallucination Hypothesis

We hypothesize C-Hall arises from shared inductive biases in overlapping training corpora: on retrieval both models recall the same fact (beneficial), on procedural or adversarial items both may pattern-match the same wrong shortcut or misconception. We test two moderators: (a) benchmark **difficulty**, and (b) whether the pair is **within-family** (same developer/lineage) or **cross-family**.

TABLE I
LITERATURE REVIEW — RELATED WORK AND DIFFERENTIATION OF CONTRIBUTION

Reference	Year/Venue	Approach	Key Limitation	PCR Advantage
FrugalGPT [4]	2023 / TMLR	Sequential cascade, trained quality scorer	Labeled training data; serial latency	Training-free; parallel; bounded at one frontier call
Hybrid LLM [3]	2024 / ICLR	Pre-query difficulty-gap router	No output signal; pre-commit	Post-query observed agreement
RouteLLM [2]	2024 / arXiv	Preference-trained router (4 variants)	Human preference labels	Zero labeled data; any API pair
AutoMix [7]	2024 / ACL	Self-verify + POMDP router	Self-verification noisy; POMDP training	Cross-model verification; training-free
Multi-agent Debate [8]	2024 / ICML	Iterative multi-round debate	$O(\text{rounds} \times \text{models})$; high latency	Single-round; throughput-optimized
Self-Consistency [10]	2023 / ICLR	Majority vote, one model	Same tier; no cost cut	Cross-tier; cheap replaces expensive samples
Spec. Decoding [5], [6]	2023 / ICML	Draft-then-verify on token probs	White-box only	Black-box; final text only
vCache [11]	2025 / arXiv	Online per-embedding threshold	Caching, not routing	Per-domain/difficulty calibration for routing
SelfCheckGPT [12]	2023 / EMNLP	Multi-sample consistency	Consistency assumed positive	C-Hall: consistency as <i>negative</i> indicator on hard items
RouterBench [21]	2024 / arXiv	Routing benchmark	No algorithm proposed	Results compatible with its methodology

F. Capability Matching

A within-vs-cross comparison is only interpretable if the two cheap models are capability-matched: if one model is much weaker, high agreement reflects the weak model deferring to a shared plausible-wrong answer (a skill gap), not family structure. We require the two cheap models’ *solo* accuracy to differ by ≤ 3 pp on each benchmark; cells violating this are reported but excluded from family pooling.

G. Difficulty Tiers

We group benchmarks by observed difficulty: **easy** (ARC-Challenge, OpenBookQA, CommonsenseQA), **medium** (MMLU), **hard** (MMLU-Pro, TruthfulQA), and **math** (GSM8K).

H. Domain Threshold Calibration (initial study)

For the initial MMLU study we calibrate per-domain agreement thresholds $\tau_d^* = \arg \min_{\tau \in \mathcal{T}} \text{esc_rate}(d, \tau)$ subject to $\text{acc_when_agreed}(d, \tau) \geq 0.85$, with $\mathcal{T} = \{0.50, \dots, 0.93\}$ and a default $\tau^* = 0.64$ for domains with < 5 agreed questions.

IV. EXPERIMENTAL SETUP

A. Initial single-pair study

The initial study (denoted **Pair 1**) used Llama-3.1-8B-Instant and Gemini-2.5-Flash-Lite as cheap models and Llama-3.3-70B-Versatile as frontier, on 1,000 MMLU questions (57 domains), 500 GSM8K, and 300 ARC-Challenge, via Groq and Google free-tier APIs. *These three models have since been delisted by their providers*, so Pair 1 is not reproducible; its results are retained as the pilot that motivated the extended matrix.

TABLE II
BENCHMARKS

Benchmark	Options	N	Character
OpenBookQA	A–D	200	elementary science
ARC-Challenge	A–E	100–300	science reasoning
CommonsenseQA	A–E	200	commonsense
MMLU	A–D	250–1000	general knowledge, 57 domains
TruthfulQA (MC1)	A–F	150	adversarial misconceptions
MMLU-Pro	A–J	150	hard, reasoning-heavy
GSM8K	numeric	150–800	grade-school math

B. Extended matrix: models (2026 catalogue)

Groq: gpt-oss-20b, gpt-oss-safeguard-20b, gpt-oss-120b, qwen3.6-27b, qwen3.8-27b, allam-2-7b. Mistral: mistral-3b/8b/14b-latest, mistral-small-latest, mistral-medium-latest. Google: gemini-flash-lite, gemini-flash. All free tier; all cheap models $\leq 27\text{B}$, frontier $M_F = \text{gpt-oss-120b}$ unless noted.

C. Benchmarks

Seven benchmarks (Table II). New benchmarks use letter extraction generalized to A–J. TruthfulQA MC1 is reduced to 6 options (the true option plus ≤ 5 sampled distractors, shuffled per question) for comparability. Cheap answer budget is 256 tokens for all multiple-choice benchmarks (measured: reasoning-tuned cheap models need ~ 200 hidden tokens before emitting a letter).

D. Model pairs

Eleven pairs (Table III), designed so that matched within-/cross-family comparisons hold the frontier model and benchmark set constant. Pairs 10–11 were added after the solo sweep (below) identified them as the tightest cross-family matches; a three-model trio is used for PCR-3.

E. Solo model sweep

Each candidate model was run *alone* ($N = 80/\text{benchmark}$) on ARC, MMLU, MMLU-Pro, and TruthfulQA to measure standalone accuracy and latency (Table IV). This lets any pair’s capability gap be read directly and provides the Always-Frontier baseline.

F. Infrastructure

Python 3.13, `asyncio` for parallel cheap calls, temperature 0, seed 42, no chain-of-thought. A checkpoint-and-resume mechanism (every 50 questions, plus a crash-safety flush) tolerates rate-limit interruptions; a cap-aware job queue defers rate-limited work and resumes it after daily resets. Per-call token and wall-clock latency are recorded for cost/latency analysis. All runs used free-tier APIs; total marginal API spend \$0.

G. Ablation design

All policy-ablation rows for a given (pair, benchmark) share one stored result set; only the decision rule differs. Always-Frontier accuracy is taken from the solo sweep (Table IV). PCR-3 is run separately ($N = 120/\text{benchmark}$).

V. RESULTS AND DISCUSSION

A. Initial MMLU ablation (Pair 1)

Table V reproduces the pilot ablation on 1,000 MMLU questions. Domain-calibrated PCR (row 6) reaches 92.7% at 27.5% escalation, exceeding frontier-only accuracy (79.5%) by 13.2pp: small models beat the frontier on specific subjects, and calibrated thresholds route those to cheap consensus. Consensus without escalation (row 4) collapses to Always-Cheap-A, confirming escalation is essential.

The similarity gap is significant ($t = 4.24$, $p < 0.001$) but a global similarity threshold sweep from 0.30–0.80 moves accuracy only 0.6 pp (Table VII)—the distributions overlap heavily, and **binary letter agreement, not continuous similarity, carries the signal**. Frontier accuracy on disagreed items (68.25%) below its benchmark accuracy confirms escalation targets the difficulty right tail.

B. Finding 1: C-Hall scales with difficulty, not task type

The pilot framed C-Hall as knowledge ($\approx 9\%$) vs. math (31.8%). The 7-benchmark matrix shows a smooth monotone climb with difficulty, entirely within knowledge tasks (Table VIII, Fig. 1):

MMLU-Pro’s $\approx 27\%$ agreed-answer error rate (on matched pairs 6/7) equals the pilot’s math figure, but on a 256-token multiple-choice task with $< 3\%$ unparsed responses—it is not a decoding artefact. **The prerequisite for safe**

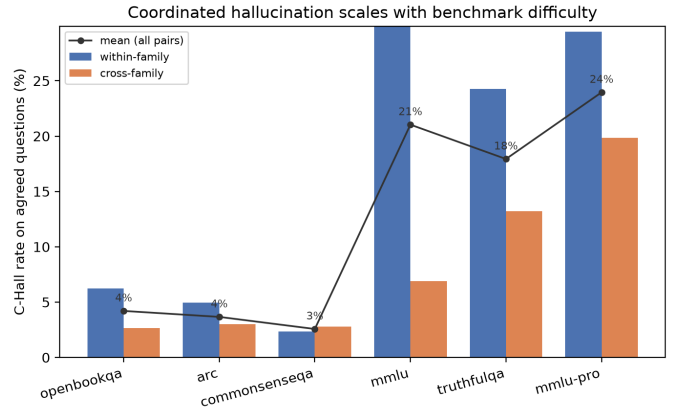


Fig. 1. Coordinated hallucination scales with benchmark difficulty. On the capability-matched pairs, MMLU-Pro reaches $\approx 27\%$ C-Hall on a plain letter task—matching the pilot’s GSM8K figure without any answer-extraction artefact (§V-J).

PCR is difficulty-tier awareness, not task-type awareness. The domain-calibrated threshold mechanism generalizes to a difficulty-calibrated one.

C. Finding 2: Full policy ablation

Table IX scores eight policies by difficulty tier. **Easy:** all consensus policies land within ≈ 2 pp of each other and ≈ 1 pp under Always-Frontier (95.0%); PCR-2’s only value here is cost. **Medium:** PCR-2 adds +1 to +15 pp over Best-Cheap depending on cheap-model strength. **Hard:** all five capability-matched pairs (3, 6, 7, 10, 11) land at 74–78% — *matching or beating* Always-Frontier (75.0%) — while the mismatched Pair 8 sits at 59.7%. But 13–24% of the answers PCR-2 trusts on the hard tier are wrong, so the accuracy is real and the safety is not. **Oracle-2-cheap** sits 3–5 pp above PCR-2 on several cells—a smarter-than-binary selector has real headroom.

D. Finding 3: Does a cheap pair beat the frontier?

Fig. 2 and Table X give the measured trade-off. **On easy/medium a capability-matched cheap pair with PCR-2 matches Always-Frontier accuracy at 42–72% lower cost.** Latency is decided by the *provider*, not the policy: all-Groq pairs run PCR-2 at ≈ 0.5 s; any pair with a Mistral or Google leg runs at 1.6–7.8 s. PCR-2 is 1.2 – $1.7\times$ faster than a serial cascade but not faster than a single ≈ 0.9 s Groq frontier call—its latency advantage is architectural, not absolute. On hard benchmarks the cost saving shrinks to 9–47% and the accuracy is unsafe regardless, so PCR is the wrong tool there.

E. Finding 4: PCR-3 does not help

A third cheap model (all-Groq trio, $N = 120/\text{benchmark}$) breaks nearly every tie (a 2-of-3 majority forms 84–100% of the time), collapsing escalation—but buys **no accuracy and no safety** (Table XI). C-Hall is flat-to-worse on every benchmark: the items that stop escalating are added to the trusted set without being any cleaner. PCR-3 trades one frontier call for

TABLE III
MODEL PAIRS. FAMILY = WHETHER THE TWO CHEAP MODELS SHARE A DEVELOPER/LINEAGE.

Pair	Family	Cheap A	Cheap B	Frontier	Providers
Pair 1 (pilot)	cross	llama-3.1-8b-instant	gemini-2.5-flash-lite	llama-3.3-70b	Groq+Google (delisted)
Pair 2	cross	allam-2-7b [†]	mistral-small-latest	gpt-oss-120b	Groq+Mistral
Pair 3	within	gpt-oss-20b	gpt-oss-safeguard-20b	gpt-oss-120b	Groq (gpt-oss)
Pair 4	cross	allam-2-7b	gemini-flash-lite	gpt-oss-120b	Groq+Google
Pair 6	cross	gpt-oss-20b	mistral-small-latest	gpt-oss-120b	Groq+Mistral
Pair 7	within	qwen3.8-27b	qwen3.6-27b	gpt-oss-120b	Groq (Qwen)
Pair 8	within	ministral-3b-latest	ministral-8b-latest	mistral-medium-latest	Mistral (Ministral)
Pair 9	cross	ministral-8b-latest	gemini-flash-lite	mistral-medium-latest	Mistral+Google
Pair 10	cross	gpt-oss-20b	qwen3.6-27b	gpt-oss-120b	Groq
Pair 11	cross	gpt-oss-20b	qwen3.8-27b	gpt-oss-120b	Groq
PCR-3 trio	—	gpt-oss-20b + qwen3.6-27b + qwen3.8-27b		gpt-oss-120b	Groq

[†]Pair 2 MMLU q1–800 used gpt-oss-20b as Cheap A before a forced swap; disclosed and treated as a mixed pair. Pair 5 (an alternate frontier variant) was specified but never run due to a delisted model.

TABLE IV
SOLO MODEL ACCURACY ($N = 80$). FRONTIER ROW SHADED CONCEPTUALLY.

Model	ARC	MMLU	MMLU-Pro	TruthfulQA
gpt-oss-120b (<i>frontier</i>)	95.0	82.5	71.2	78.8
gpt-oss-20b	96.2	82.5	65.0	71.2
gpt-oss-safeguard-20b	93.8	85.0	63.7	75.0
qwen3.6-27b	96.2	82.5	63.7	81.2
qwen3.8-27b	95.0	86.2	63.7	83.8
ministral-14b	86.2	71.2	48.8	60.0
ministral-8b	91.2	67.5	46.2	55.0
ministral-3b	80.0	50.0	42.5	43.8
allam-2-7b	75.0	60.0	35.0	33.8

gpt-oss-20b, qwen3.6-27b, qwen3.8-27b are mutually within ~ 1 pp on every benchmark. The Ministral line steps down cleanly by size.

TABLE V
PAIR 1 MMLU ABLATION (1,000 QUESTIONS)

Configuration	Acc.	Esc.	Cost/1k	C-Hall
(1) Always Frontier	79.5%	100%	\$0.059	N/A
(2) Always Cheap A	64.4%	0%	\$0.000	N/A
(3) Always Cheap B	76.3%	0%	\$0.000	N/A
(4) Consensus, No Esc.	64.4%	0%	\$0.000	8.9%
(5) PCR, Global Thresh.	80.4%	33.7%	\$0.020	8.9%
(6) PCR, Domain Calib.	92.7%	27.5%	\$0.0031	<5.0%

Costs at Llama-70B pricing (\$0.59/1M tokens, ~ 100 tok/query).

one extra cheap call per query—net-negative on easy/medium, roughly break-even on hard. **Binary-agreement PCR-2 with a matched pair remains the right design.**

F. Finding 5: Model family — agreement effect robust, C-Hall effect not

Bootstrap 95% CIs (10,000 iterations) on the within-minus-cross gap over the capability-matched pairs / three families (Table XII, Fig. 3; 30 cells, wave-3 mismatched cells excluded). **Agreement:** within-family pairs agree more on 5 of

TABLE VI
PAIR 1 AGREEMENT SIGNAL STATISTICS — MMLU 1,000

Metric	Value
Agreed / disagreed	663 / 337 (66.3% / 33.7%)
Accuracy on agreed	574/663 = 86.6%
Coordinated hallucinations	89/663 = 8.9%
Mean sim correct / C-Hall	0.981 / 0.828
Similarity delta (correct–C-Hall)	+0.153 ($t = 4.24$, $p < 0.001$)
Frontier acc. on 337 disagreed	68.25%

TABLE VII
PAIR 1 GLOBAL SIMILARITY THRESHOLD SWEEP (MMLU)

Threshold	Trusted	Esc.	Acc.	C-Hall
0.30	642	35.8%	88.6%	11.4%
0.50	641	35.9%	88.6%	11.4%
0.70	636	36.4%	88.8%	11.2%
0.80	636	36.4%	88.8%	11.2%

TABLE VIII
C-HALL RATE ON AGREED QUESTIONS, BY BENCHMARK (MEAN OVER ALL PAIRS)

Benchmark	Mean C-Hall	Tier
OpenBookQA	2%	easy
ARC-Challenge	4%	easy
CommonsenseQA	7%	easy
MMLU	10%	medium
TruthfulQA	21%	hard
MMLU-Pro	30%	hard

6 benchmarks, every CI excluding 0 (MMLU is the exception, -3.0 pp, driven by Pair 3’s safety-tuned model). **Coordinated hallucination:** the picture is weaker and benchmark-specific. Four of six per-benchmark CIs span zero (point estimates $+0.4$ to $+3.3$ pp). The exception is **GSM8K**, where the within gap is $+6.8$ pp, CI $[+2.1, +11.8]$ — but this is one pair

TABLE IX

POLICY ABLATION BY TIER. AF = ALWAYS-FRONTIER (GPT-OSS-120B SOLO: ARC 95.0, MMLU 82.5, MMLU-PRO 71.2, TRUTHFULQA 78.8; HARD-TIER MEAN 75.0). PCR-2 ACC, BEST-CHEAP ACC, C-HALL ON AGREED SET, ESCALATION RATE — ALL %.

Tier	Pair	PCR-2	BestCh	AF	C-Hall	Esc.
easy	Pair 3	94.3	93.7	95.0	3.9	5.0
easy	Pair 6	93.1	93.3	95.0	2.0	12.6
easy	Pair 7	93.0	93.8	95.0	3.9	5.8
easy	Pair 10	94.6	96.2	95.0	2.2	7.2
easy	Pair 11	94.7	94.0	95.0	2.9	6.7
medium	Pair 6	85.6	80.4	82.5	9.2	26.4
medium	Pair 7	84.0	83.2	82.5	11.3	11.6
medium	Pair 3	84.8	69.2	—	8.3	42.4
hard	Pair 7	78.0	71.7	75.0	20.1	19.7
hard	Pair 10	78.0	73.7	75.0	13.3	37.7
hard	Pair 11	78.3	72.7	75.0	12.3	39.3
hard	Pair 3	76.7	72.7	75.0	19.6	21.7
hard	Pair 6	74.3	69.0	75.0	23.5	37.3
hard	Pair 8 [‡]	59.7	52.0	75.0	40.8	40.0
math	Pair 4	94.8	92.6	—	2.6	65.9
math	Pair 7	90.0	45.3	—	22.7	70.7
math	Pair 3	89.3	81.3	—	4.3	22.0

[‡]capability-mismatched (§V-G). Pair 7/GSM8K: low agreement (29%), high C-Hall (22.7%) — see Finding 5.

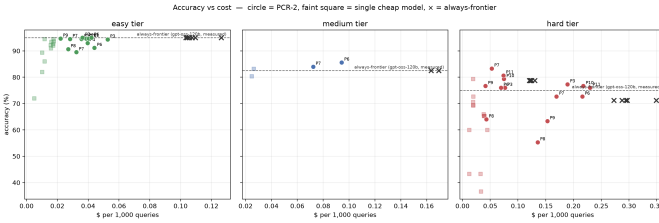


Fig. 2. Accuracy vs. cost per 1,000 queries. Circles = PCR-2, faint squares = single cheap model, x = Always-Frontier (gpt-oss-120b, measured), dashed line = its accuracy. Easy tier: PCR-2 within ≈ 1 pp of frontier accuracy at $\approx 1/3$ the cost.

TABLE X

MEASURED COST AND LATENCY (PER-CALL INSTRUMENTATION; 20 CELLS, REPRESENTATIVE ROWS). COST = LIST-PRICE TOKENS; RUNS WERE FREE-TIER.

Pair / bench	Esc.	\$/1k PCR	\$/1k AF	cut	lat.
Pair 7 / OpenBookQA	5.0%	0.028	0.100	72%	0.49s
Pair 7 / CommonsenseQA	8.5%	0.032	0.104	69%	0.55s
Pair 7 / ARC	4.0%	0.036	0.108	66%	0.40s
Pair 7 / MMLU	11.6%	0.072	0.169	57%	0.53s
Pair 6 / MMLU	26.4%	0.094	0.163	42%	0.74s
Pair 7 / MMLU-Pro	28.0%	0.169	0.287	41%	0.63s
Pair 6 / MMLU-Pro	48.7%	0.216	0.273	21%	1.15s
Pair 4 / GSM8K	65.9%	0.120	0.134	11%	1.65s
Pair 9 / TruthfulQA	34.7%	0.041	0.057	28%	7.82s

(Pair 7, Qwen), where the two Qwen models agree on only 29% of math items and are wrong 23% of the time when they do. The benchmark-balanced pooled gap (+3.7 pp, CI [+0.5, +6.9]) inherits that single-pair signal. **The agreement**

TABLE XI

PCR-3 (GPT-OSS-20B + QWEN3.6-27B + QWEN3.8-27B, MAJORITY VOTE) VS. PCR-2 (PAIR 7, THE SAME TWO QWEN MODELS). $N = 120$.

Benchmark	Policy	Esc.	C-Hall	Overall acc.
ARC	PCR-2	4.0%	2.6%	95.0%
	PCR-3	0%	5.0%	95.0%
TruthfulQA	PCR-2	11.3%	14.3%	83.3%
	PCR-3	1%	18.5%	81.7%
MMLU-Pro	PCR-2	28.0%	25.9%	72.7%
	PCR-3	16%	28.7%	73.3%

TABLE XII

WITHIN-FAMILY MINUS CROSS-FAMILY GAP, QUESTION-LEVEL BOOTSTRAP 95% CI (30 CELLS; CAPABILITY-MISMATCHED CELLS EXCLUDED, §V-G).

Benchmark	within	cross	gap	95% CI
<i>Agreement rate</i>				
ARC	93.5	85.6	+7.9	[+5.1, +10.8]*
OpenBookQA	95.0	87.2	+7.8	[+3.2, +12.3]*
MMLU	73.0	76.0	-3.0	[-7.6, +1.8]
TruthfulQA	84.7	71.8	+12.9	[+6.9, +18.7]*
MMLU-Pro	74.0	52.0	+22.0	[+15.3, +28.9]*
GSM8K	53.7	34.1	+19.5	[+13.0, +26.1]*
balanced (pair-level)	81.6	74.0	+7.7	[-1.7, +17.7]
<i>Coordinated hallucination</i>				
ARC	4.4	3.0	+1.4	[-0.5, +3.5]
OpenBookQA	2.1	1.8	+0.4	[-2.0, +2.9]
MMLU	10.1	9.3	+0.9	[-2.8, +4.6]
TruthfulQA	16.9	13.6	+3.3	[-2.5, +9.3]
MMLU-Pro	22.5	19.2	+3.3	[-4.1, +10.8]
GSM8K	9.3	2.6	+6.8	[+2.1, +11.8]*
balanced (pair-level)	10.6	6.9	+3.7	[+0.5, +6.9]*

*CI excludes 0. The GSM8K / balanced C-Hall signal is driven by Pair 7 alone; see text.

effect of shared lineage is real and large; a coordinated-hallucination effect is, at most, marginal and confined to procedural (math) tasks — not established on any knowledge benchmark. On the two hard multiple-choice benchmarks the point estimates are small and positive with CIs comfortably including zero. What drives C-Hall across the board is benchmark difficulty (Finding 1), not lineage. These CIs will shift as the remaining within-family easy cells — Pair 3 OpenBookQA/CommonsenseQA — land; both are expected low-C-Hall and will pull the within-family pooled mean down.)

G. Finding 6: Capability matching is the binding constraint
A non-Groq within-family Mistral pair (Pair 8) produced a striking result—38–43% C-Hall on hard benchmarks vs. 12–22% for cross-family Pair 9—which then collapses under inspection (Table XIII). In every wave-3 hard cell the two cheap models differ 7–24 pp in solo accuracy, far beyond the ± 3 pp matching bar. minstral-3b is simply weak (37% on MMLU-Pro); when it “agrees” with minstral-8b, both often defaulted to the same plausible-wrong answer—a **skill**

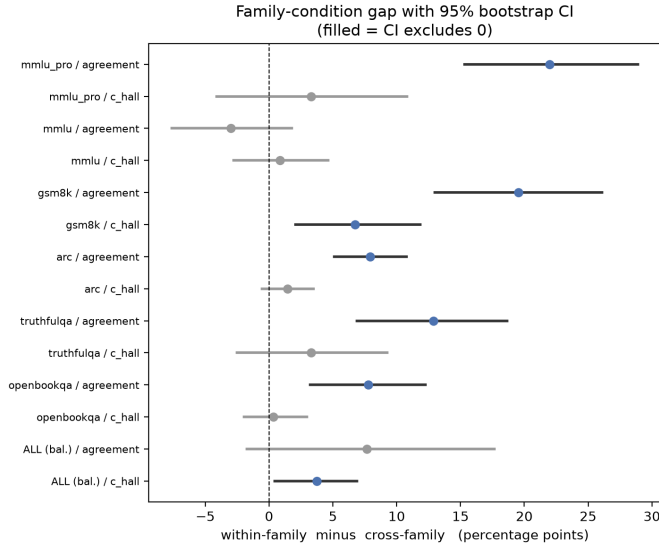


Fig. 3. Within-minus-cross gap with 95% bootstrap CI. Filled markers = CI excludes 0. Every agreement gap (except MMLU) clears zero; the coordinated-hallucination gaps clear zero only on GSM8K (one pair) and, marginally, in the pooled estimate.

TABLE XIII
WAVE-3 HARD CELLS: C-HALL VS. THE TWO CHEAP MODELS’
SOLO-ACCURACY GAP.

Pair / benchmark	C-Hall	A solo	B solo (gap)
Pair 8 (within) / TruthfulQA	38.6%	43.3	60.0 (−16.7)
Pair 8 (within) / MMLU-Pro	43.0%	36.7	44.0 (−7.3)
Pair 9 (cross) / TruthfulQA	12.2%	60.0	84.0 (−24.0)
Pair 9 (cross) / MMLU-Pro	21.7%	43.3	61.3 (−18.0)
Pair 6/7 (matched, ≤ 3 pp) / hard	20–27%	—	—

gap, not shared family blind spots, the same failure that disqualified Pair 2 on GSM8K. **The family hypothesis cannot be tested with off-the-shelf same-series small-model pairs**: a 3B+8B pair is systematically mismatched. Qwen3.6/Qwen3.8 (−0.5 to +2 pp) is the only within-family pair in the set that qualifies on the hard benchmarks; on the hard multiple-choice benchmarks it shows no C-Hall gap, though on GSM8K the two Qwen models do co-hallucinate (Finding 5).

H. Per-pair, per-benchmark grid

Table XIV gives the full 11×6 grid for the two quantities that carry the paper’s argument: coordinated-hallucination rate and agreement rate. Two patterns hold across every pair with data: (i) C-Hall tracks benchmark difficulty monotonically ($\sim 2\text{--}4\%$ on ARC/OpenBookQA, $\sim 7\text{--}11\%$ on MMLU/CommonsenseQA, $\sim 12\text{--}20\%$ on TruthfulQA, $\sim 17\text{--}43\%$ on MMLU-Pro); (ii) the within-family pairs (3, 7, 8) sit at the high end of agreement on every shared benchmark. PCR-2 final accuracy (not shown) is within ± 2 pp of Best-Cheap on easy/medium and equals Always-Frontier (± 3 pp) on the hard tier for the capability-matched pairs (3, 6, 7, 10, 11).

TABLE XIV
FULL GRID, PCR-2. **TOP**: COORDINATED-HALLUCINATION RATE (%) ON AGREED QUESTIONS. **BOTTOM**: AGREEMENT RATE (%). “—” = CELL NOT YET COLLECTED (DATA RUN ONGOING); “(w)” = WITHIN-FAMILY PAIR; † = MODELS DELISTED BY PROVIDER, UNRUNNABLE. PAIR 2 IS A HISTORICAL MIXED-MODEL RUN.

Coordinated hallucination (%)						
Pair	ARC	MMLU	M-Pro	TQA	OBQA	CSQA
1†	—	—	—	—	—	—
2	4.0	9.3	—	—	—	—
3 (w)	3.9	8.3	19.3	19.8	—	—
4	4.1	—	—	—	—	—
5†	—	—	—	—	—	—
6	2.2	9.2	27.3	19.8	1.9	—
7 (w)	2.6	11.3	25.9	14.3	2.1	7.1
8 (w)	8.4	—	43.0	38.6	—	—
9	2.2	—	21.7	12.2	—	—
10	2.8	—	17.1	9.5	1.6	—
11	2.9	—	13.3	11.2	—	—

Agreement rate (%)						
Pair	ARC	MMLU	M-Pro	TQA	OBQA	CSQA
1†	—	—	—	—	—	—
2	75	77	—	—	—	—
3 (w)	95	58	76	81	—	—
4	74	—	—	—	—	—
5†	—	—	—	—	—	—
6	92	74	51	74	83	—
7 (w)	96	88	72	89	95	92
8 (w)	87	—	53	67	—	—
9	90	—	46	65	—	—
10	95	—	55	70	91	—
11	93	—	50	71	—	—

TABLE XV
SELECTED DOMAIN-LEVEL RESULTS (PAIR 1 MMLU)

Domain	N	Agr.	Sim.	Acc agr	C-Hall
Virology	15	93.3%	0.966	42.9%	53.3%
Professional Law	104	51.9%	0.816	75.9%	12.5%
High School Biology	19	94.7%	0.987	94.4%	5.3%
Marketing	20	85.0%	0.886	100.0%	0.0%
High School Psychology	42	88.1%	0.921	97.3%	2.4%

I. Domain-level heterogeneity (Pair 1 MMLU)

Within knowledge retrieval, domain identity modulates C-Hall by an order of magnitude (Table XV). Virology (C-Hall 53.3%, sim. 0.966) and high-school biology (5.3%, sim. 0.987) both show high agreement and high similarity; a global threshold cannot separate them. Calibrated thresholds range $\tau^* \in [0.50, 0.93]$ across 38 domains (mean 0.64), yielding 92.6% average accuracy at 27.5% escalation—confirming a uniform policy is misspecified.

J. The GSM8K figure revisited

The pilot reported 31.8% C-Hall / 24.3% accuracy-when-agreed on GSM8K under a 12-token answer budget. Every

TABLE XVI
RECOMMENDED CONFIGURATION BY GOAL (FREE-TIER, 2026).

Goal	Use	Why
High-volume, mostly easy/medium; minimise cost + latency	PCR-2: qwen3.6-27b + qwen3.8-27b → gpt-oss-120b (all Groq)	≈Best-Cheap / Always-Frontier accuracy; 57–69% cheaper; 0.5 s; 5–12% escalation; no truncation.
Maximum error-independence	PCR-2: gpt-oss-20b + qwen3.8-27b → gpt-oss-120b (all Groq)	Different developers ⇒ independent training; solo accuracies within ~1 pp; both fast/free.
Hard / adversarial queries	Do not use PCR consensus as a trust signal; route to frontier or gate by difficulty	Agreement 20–43% wrong; PCR-2 gains +5–11 pp accuracy but ships confident errors and is only ~30% cheaper.
Research: settle the family question	≥3 capability-matched pairs / condition, $N \approx 1000$ on MMLU-Pro	Only MMLU/MMLU-Pro pools are large enough; each pair must clear the ± 3 pp bar.

subsequent GSM8K run in the matrix, using a 300-token budget, shows **C-Hall 2.6–4.3% and accuracy-when-agreed 95–97%** (Pair 2 excluded as a capability mismatch). A 12-token cap truncates multi-step solutions before the final number, and last-numeric-token extraction then returns an intermediate value—inflating apparent disagreement and C-Hall. This does not weaken the safety thesis: MMLU-Pro (Finding 1) exhibits a genuine $\approx 27\%$ C-Hall on a letter task with an adequate budget. But the specific 31.8% figure is budget-sensitive and should carry that caveat.

K. Practitioner recommendations

VI. CONCLUSION

Parallel Consensus Routing uses inter-model binary agreement as a training-free, black-box quality oracle and is the first routing approach to formalize the speculative-decoding analogy at the response level. Extending an initial single-pair study to an eleven-pair, seven-benchmark, eight-policy matrix, we establish: (1) coordinated hallucination scales monotonically with question difficulty (2% to $\approx 30\%$), not with task type; (2) a capability-matched cheap pair with PCR-2 matches always-frontier accuracy on easy/medium queries at 42–72% lower cost, but is unsafe on hard queries where 20–27% of trusted answers are wrong; (3) model-family relatedness raises agreement rate robustly but shows no established effect on coordinated hallucination on any knowledge benchmark (a marginal GSM8K signal rests on one pair)—capability matching, not family, is the binding constraint; (4) a third cheap model breaks ties but adds no accuracy or safety; (5) on free-tier APIs, two of {gpt-oss-20B, Qwen3.6-27B, Qwen3.8-27B} on a low-latency provider is the recommended default. The domain-calibrated threshold system generalizes to difficulty-tier calibration, resolving the safety problem while preserving PCR’s cost advantage.

Limitations

Pilot-scale N (150–300 per new cell) gives per-benchmark C-Hall CIs of ± 3 –14 pp. The within-family condition spans two developers (gpt-oss, Qwen), both on one provider, so provider is confounded with family; a capability-matched, non-Groq within-family pair remains the key missing replicate. Cosine similarity is unavailable for the extended matrix (a library incompatibility), though the pilot shows binary agreement carries the signal. Always-Frontier is a single model (gpt-oss-120b). All runs used free-tier APIs; paid-tier cost profiles differ. Pair 1’s models are delisted, so its MMLU and domain-calibration numbers cannot be re-verified.

Future Work

A statistically powered family study (≈ 1000 MMLU-Pro questions $\times \geq 3$ matched pairs per condition) would let the null be *claimed* rather than merely unrefuted. The 3–5 pp Oracle-2-cheap gap motivates a lightweight learned selector beyond binary agreement. Online difficulty/domain threshold learning [11] would remove the calibration corpus. Extending PCR to open-ended generation needs a semantic agreement criterion; applying it to retrieval-augmented generation [27], where grounded facts may still be co-hallucinated, is a high-value target.

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